Lecture 1

Introduction to Data Structures

CSEN 1038 Advanced Data Structures and Algorithms

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What is a Data Structure?

What is a Data Structure?

A data structure is a container of data that structures the data in a way that allows specified queries.

Most of the programming languages support built-in data strcutures, like arrays, sets, maps, etc...

The world of queries problems

The world of queries problems

In the realm of database systems, the intersection of querying and efficiency is vital. These systems are designed to manage massive datasets while ensuring that information can be retrieved almost instantaneously.

Structure of a Query Problem

A standard query-based problem generally follows a three-step lifecycle:

- **Initialization:** The starting state or the initial dataset provided to the system.
- **Update Operations:** The ability to modify, add, or delete data dynamically.
- **Query Execution:** The mechanism for retrieving specific information or calculating results based on the current data.

Subarray sum

Subarray sum: Problem Framework

Applying the standard query-based structure to this problem looks like this:

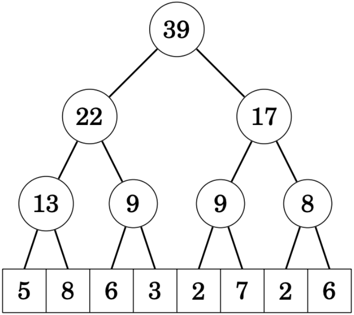
1. **Initialization:** You are given an initial array of n numerical elements, often represented as $A = [a_1, a_2, \dots, a_n]$.
2. **Support Updates:** The system must handle changes to the data, such as updating the value of an element at a specific index i (Point Updates) or modifying a range of values (Range Updates).
3. **Support Queries:** The core task is to quickly calculate the sum of elements within a given range $[L, R]$, defined as:

$$\sum_{i=L}^{R} a_i$$

Segment tree structure

Segment tree structure

For example, the segment tree



is stored as follows:

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
39	22	17	13	9	9	8	5	8	6	3	2	7	2	6

More problems

More problems

Other problems include:

1. Maximum / Minimum in a range
2. Bitwise operations like AND, OR, and XOR
3. Maximum subarray sum
4. Progressive sum queries

Kth one problem

Kth one Problem: Framework

This problem follows the standard query-update lifecycle to ensure high performance even as the bitmask changes:

1. **Initialization:** You are provided with an initial array of length n consisting entirely of binary digits (0s and 1s).
2. **Support Updates:** The system must handle bit-flips. For example, changing a 0 to a 1 at index i, or vice versa, effectively shifting the positions of all subsequent ones.
3. **Support Queries:** Given an integer k, the system must return the index of the k-th occurrence of 1 in the array. If the total count of 1s is less than k, it returns an error or a null value.

Other variations

Other variations of segment trees

The segment tree discussed is the basic type, usually called *Point update, Range query*.

Other variations of segment trees are:

1. Range update, Point query
2. Range update, Range query

Questions?

References i

- https://cp-algorithms.com/data_structures/segment_tree.html
- https://www.youtube.com/watch?v=xRiKTCG2IVg&list=PLc02D4EoVYQDIdPLUC0zXHUWUVXj7C7Qi&index=9